

BARRIER METHODS

Lecture 29: Presented by Jacomine Grobler



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In this chapter, we again consider non-linear optimisation problems in which the objective is to

$$\text{maximise } z = f(\mathbf{x})$$

subject to a set of constraints of the form

$$\begin{aligned}g_j(\mathbf{x}) &\leq a_j, & j = 1, \dots, p, \\h_k(\mathbf{x}) &= b_k, & k = 1, \dots, q, \\ \mathbf{x} &\in \mathbb{R}^n,\end{aligned}$$

where p and q are any non-negative integers, n is a natural number, the functions $f, g_1, \dots, g_p$ and $h_1, \dots, h_q$ are all twice differentiable with respect to all pairs of their argument components, and where $a_1, \dots, a_p$ and $b_1, \dots, b_q$ are real constants.

Barrier methods are iterative methods for approximating optimal solutions to problem instances in which there are only inequality constraints.

We generate a sequence of approximates which lie strictly within the feasible region, denoted here by $\mathcal{X}$.

Feasibility is maintained during the generation of the sequence of barrier method approximates by creating a barrier that keeps these approximates away from the boundary of the feasible region.

This barrier is created by the introduction of a function $\phi : \mathcal{X} \mapsto \mathbb{R}$ which is continuous in the interior of $\mathcal{X}$ and which grows without bound as the boundary of $\mathcal{X}$ is approached from within $\mathcal{X}$.

That is,

$$\phi(\boldsymbol{x}) \rightarrow \infty \quad \text{as} \quad g_j(\boldsymbol{x}) \rightarrow a_j^-.$$

Two popular examples of such functions are the logarithmic function

$$\phi(\mathbf{x}) = \sum_{j=1}^p \log[a_j - g_j(\mathbf{x})]$$

and the inverse function

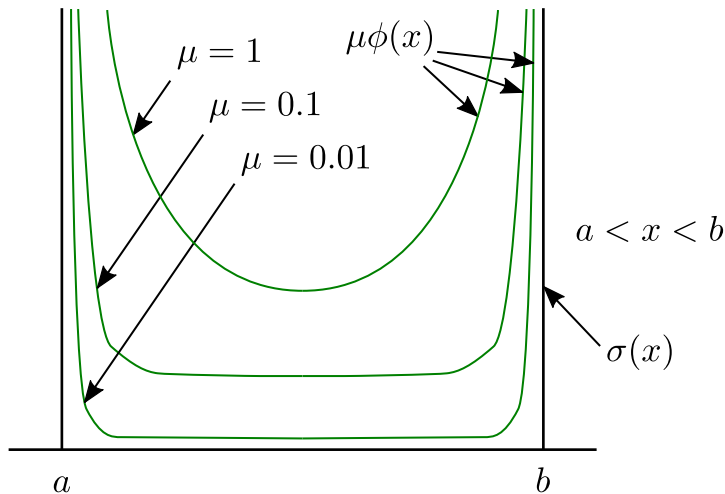
$$\phi(\mathbf{x}) = - \sum_{j=1}^p \frac{1}{a_j - g_j(\mathbf{x})}.$$

In both these cases, the scaled function $\mu\phi(\mathbf{x})$ approaches a *barrier* $\sigma : \mathbb{R}^n \mapsto \mathbb{R}$ defined by

$$\sigma(\mathbf{x}) = \begin{cases} 0 & \text{if } \mathbf{x} \in \mathcal{X} \\ \infty & \text{if } \mathbf{x} \notin \mathcal{X}, \end{cases}$$

as $\mu \rightarrow 0^+$.

Barrier methods



Barrier methods

By adding a *barrier term* $\mu\phi(\mathbf{x})$ to the objective function $f(\mathbf{x})$ in order to obtain a so-called *barrier function*

$$\beta(\mathbf{x}; \mu) = f(\mathbf{x}) + \mu\phi(\mathbf{x}),$$

it therefore follows that the constrained optimisation problem may be solved by maximising the (unconstrained) barrier function under the condition that $\mu \rightarrow 0^+$. It follows from the discussion above that two ubiquitous barrier functions are the *logarithmic barrier function*

$$\beta(\mathbf{x}; \mu) = f(\mathbf{x}) + \mu \sum_{j=1}^p \log[a_j - g_j(\mathbf{x})]$$

and the *inverse barrier function*

$$\beta(\mathbf{x}; \mu) = f(\mathbf{x}) - \mu \sum_{j=1}^p \frac{1}{a_j - g_j(\mathbf{x})}.$$

Maximise

$$f(x) = -0.1x^2 + 10$$

subject to the constraint

$$x \leq 10.$$

The logarithmic barrier function is

$$\beta(x) = -0.1x^2 + 10 + \mu \log(10 - x)$$

and the inverse barrier function is

$$\beta(x) = -0.1x^2 + 10 - \mu \frac{1}{10 - x}.$$

Algorithm 6.1 (Barrier Method)

Input: An instance $\mathcal{I}$ of (6.1)–(6.4) in which $q = 0$, an initial estimate $\mathbf{x}^0$ as to an optimal solution to $\mathcal{I}$, an initial barrier parameter value μ^0 , a barrier parameter reduction factor $\delta \in (0, 1)$, and a search termination criterion

Output: An approximation of an optimal solution to $\mathcal{I}$ satisfying the search termination criterion

Step 1 $k \leftarrow 0$

Step 2 If the search termination criterion is met, then output $\mathbf{x}^k$ and stop

Step 3 Taking $\mathbf{x}^k$ as starting point, maximise the barrier function in (6.6) for the fixed parameter value $\mu = \mu^k$ by invoking any numerical method of Chapter 3

Step 4 $\mu^{k+1} = \delta\mu^k$, $k \leftarrow k + 1$

Step 5 Return to Step 2

Example 6.1 (pp. 181–182)

Consider the problem of

$$\text{maximising } z = 2x_2 - x_1$$

subject to the constraints

$$\begin{aligned}x_2^2 - x_1 &\leq 1, \\x_2 &\geq 0, \\(x_1, x_2) &\in \mathbb{R}^2.\end{aligned}$$

Estimating Lagrange multipliers

There is an important relationship between the use of barrier methods and the Karush-Kuhn-Tucker necessary conditions for local optimality in a constrained optimisation problem.

We may therefore estimate the Lagrange multipliers as

$$\lambda_j(\mu) = \frac{\mu}{a_j - g_j(\hat{\mathbf{x}})}.$$

Let us revisit Example 6.1 to determine the Lagrange multipliers.

Advantages of barrier methods:

- They avoid the combinatorial difficulties typically associated with having to test different combinations of active constraint sets, a notable disadvantage of applying the Karush-Kuhn-Tucker conditions.
- They converge to optimal solutions of problem instances in which $q = 0$ under the sufficiency conditions of Theorem 5.2.
- They are able to provide estimates of Lagrange multipliers at optimal solutions to problem instances in which $q = 0$.
- The logarithmic and inverse barrier functions are both necessarily concave if the objective function $f(\mathbf{x})$ itself is concave.

Disadvantages of barrier methods:

- May become increasingly difficult to maximise the (unconstrained) barrier function $\beta(\mathbf{x}; \mu)$ as the value of μ decreases, because of numerical instabilities introduced when μ becomes very small.
- The difficulty of choosing an initial step length candidate $\hat{\alpha}_k^0$ when employing a backtracking line search as part of the numerical method employed in Step 3 of Algorithm 6.1 during its k -th iteration so that the backtracking process can start on the feasible side of the barrier.

Example 6.3 (pp. 187–188)

Consider the problem of

$$\text{minimising } f_{6.2}(x_1, x_2) = x_1^2 + x_2^2$$

subject to the constraints

$$x_1 \geq 1$$

$$x_2 \geq -1.$$